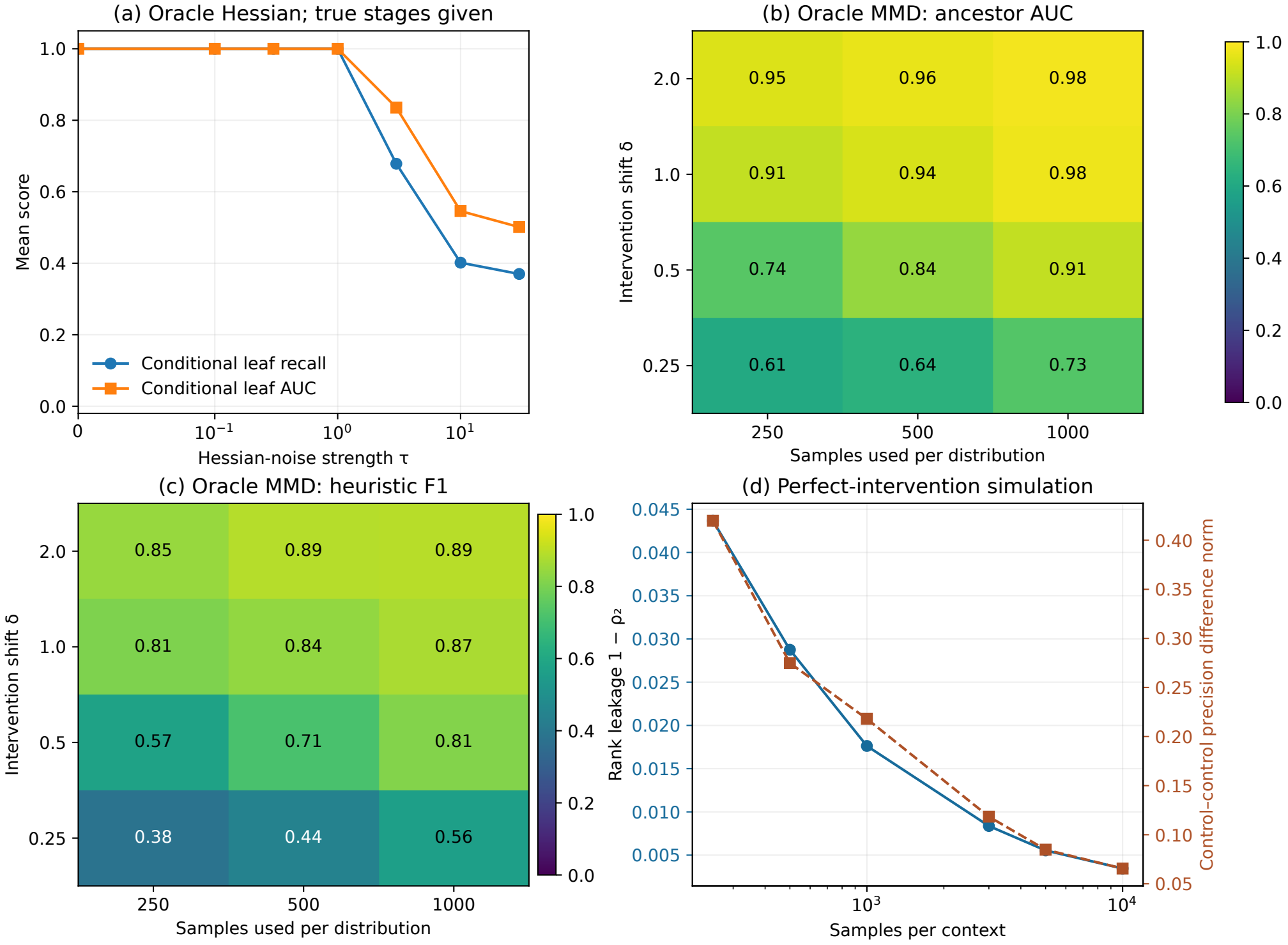
(a) Oracle Hessian; true stages given
Mean score
Hessian-noise strength τ
Conditional leaf recall
Conditional leaf AUC
(b) Oracle MMD: ancestor AUC
Intervention shift δ
Samples used per distribution
0.95 0.96 0.98
0.91 0.94 0.98
0.74 0.84 0.91
0.61 0.64 0.73
(c) Oracle MMD: heuristic F1
Intervention shift δ
Samples used per distribution
0.85 0.89 0.89
0.81 0.84 0.87
0.57 0.71 0.81
0.38 0.44 0.56
(d) Perfect-intervention simulation
Rank leakage 1 − ρ₂
Control–control precision difference norm
Samples per context